Parametric Pavilion

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Assignment 2
Arch 565-Advanced Computer Graphics II
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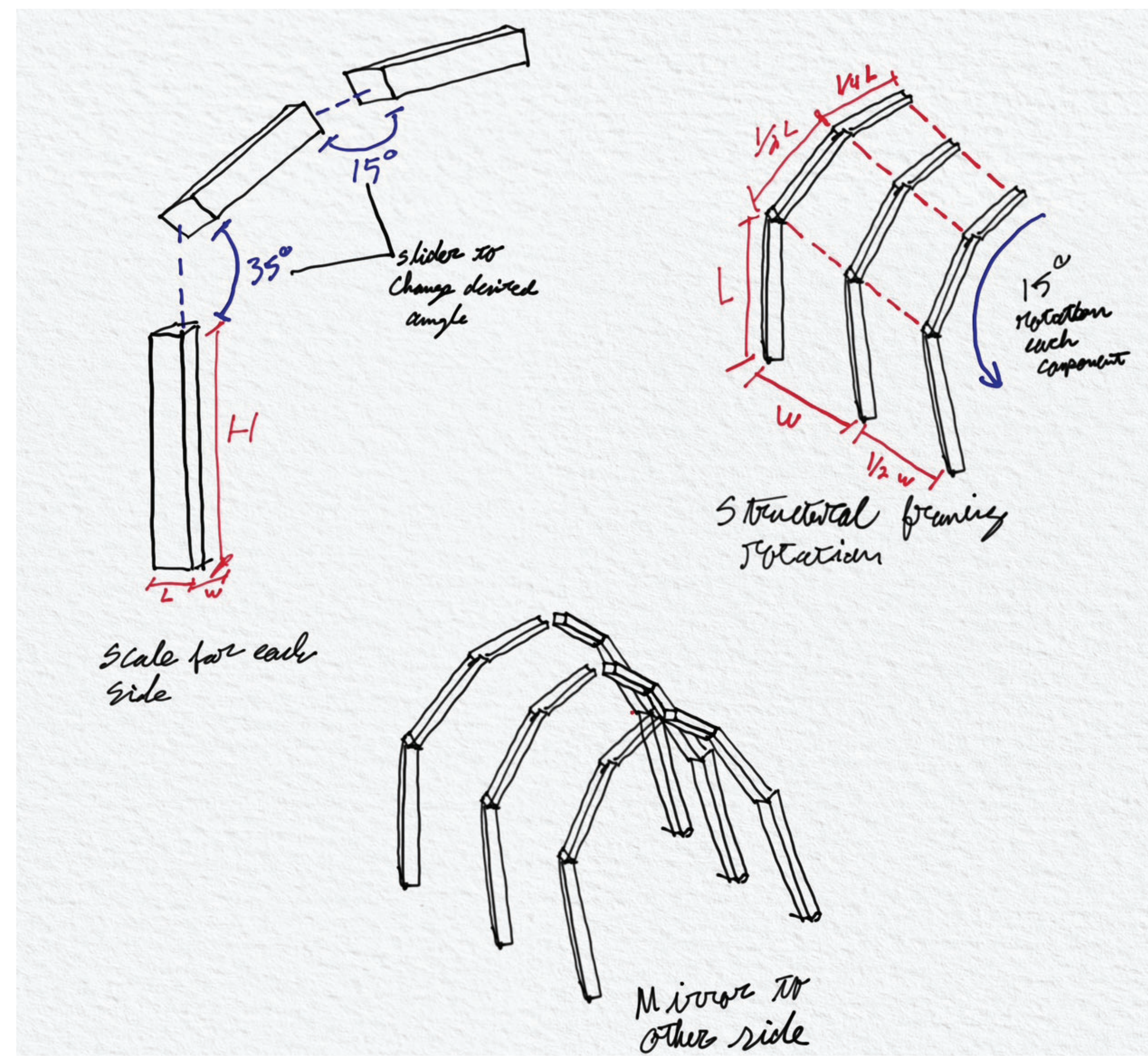
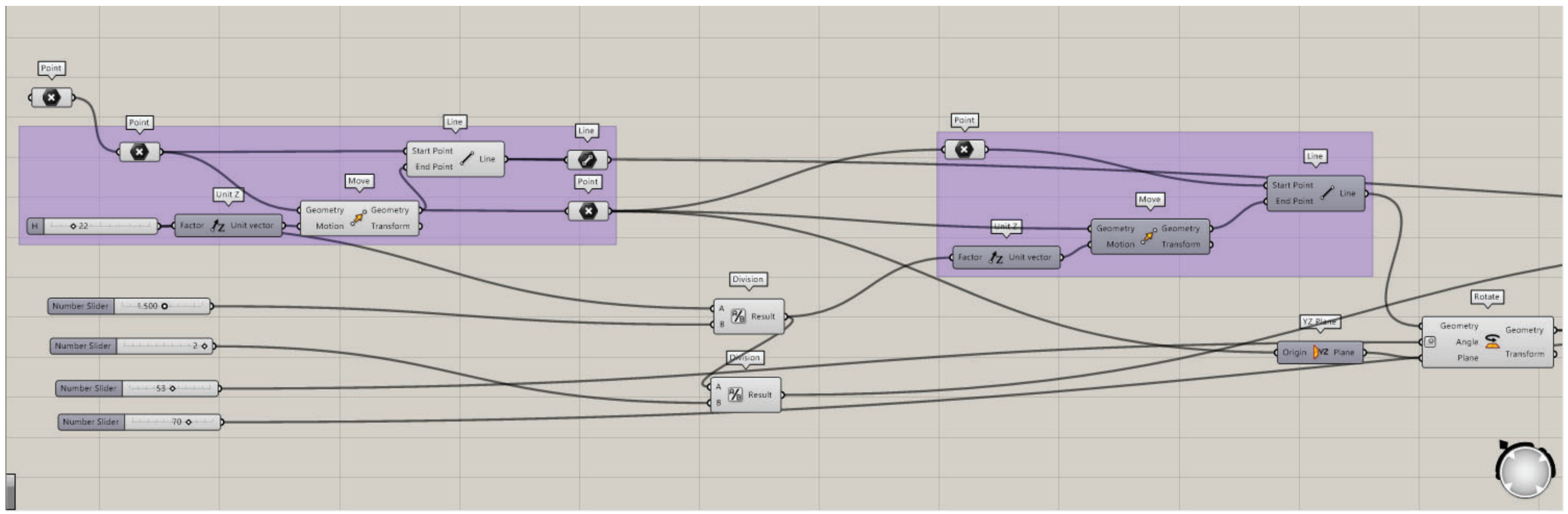
The design of the pavilion was to be more of a tunnel like form one would walk through. The concept was to have the dimensions of the segments and angle of the design be parametric. As the arch was formed each segment is half of the previous segment of the structure. The goal was to make a script for one side of the covered space and then would mirror the components to make a tunnel with the same dimensions and angles consistent through the design.

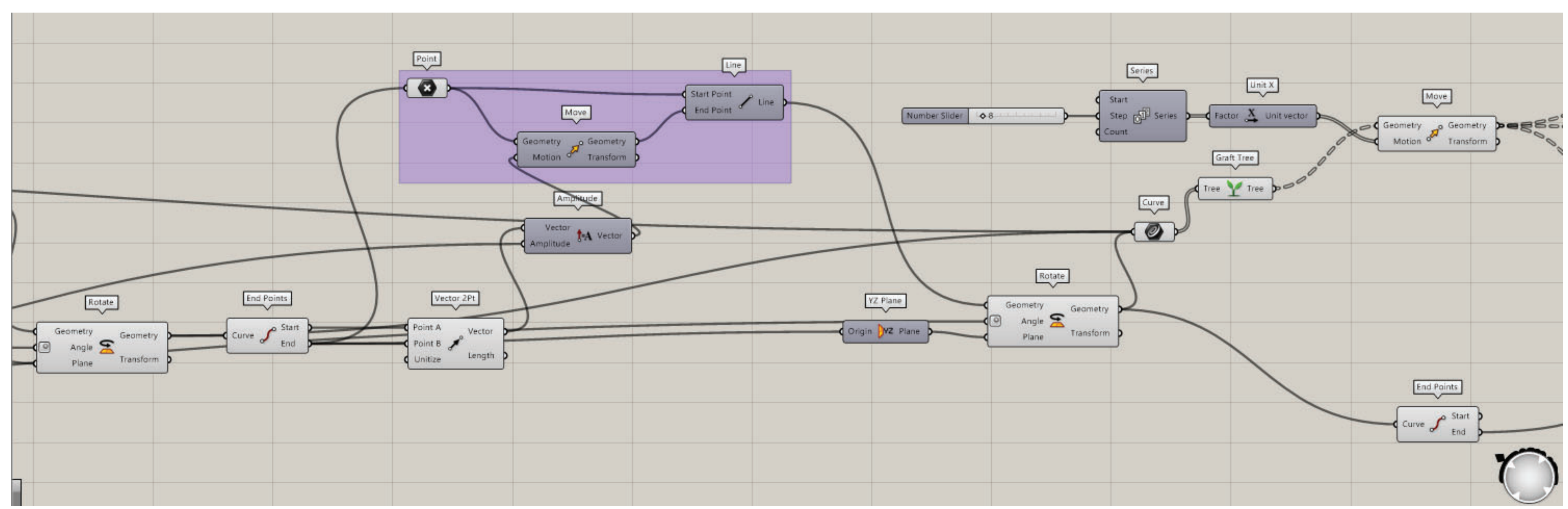


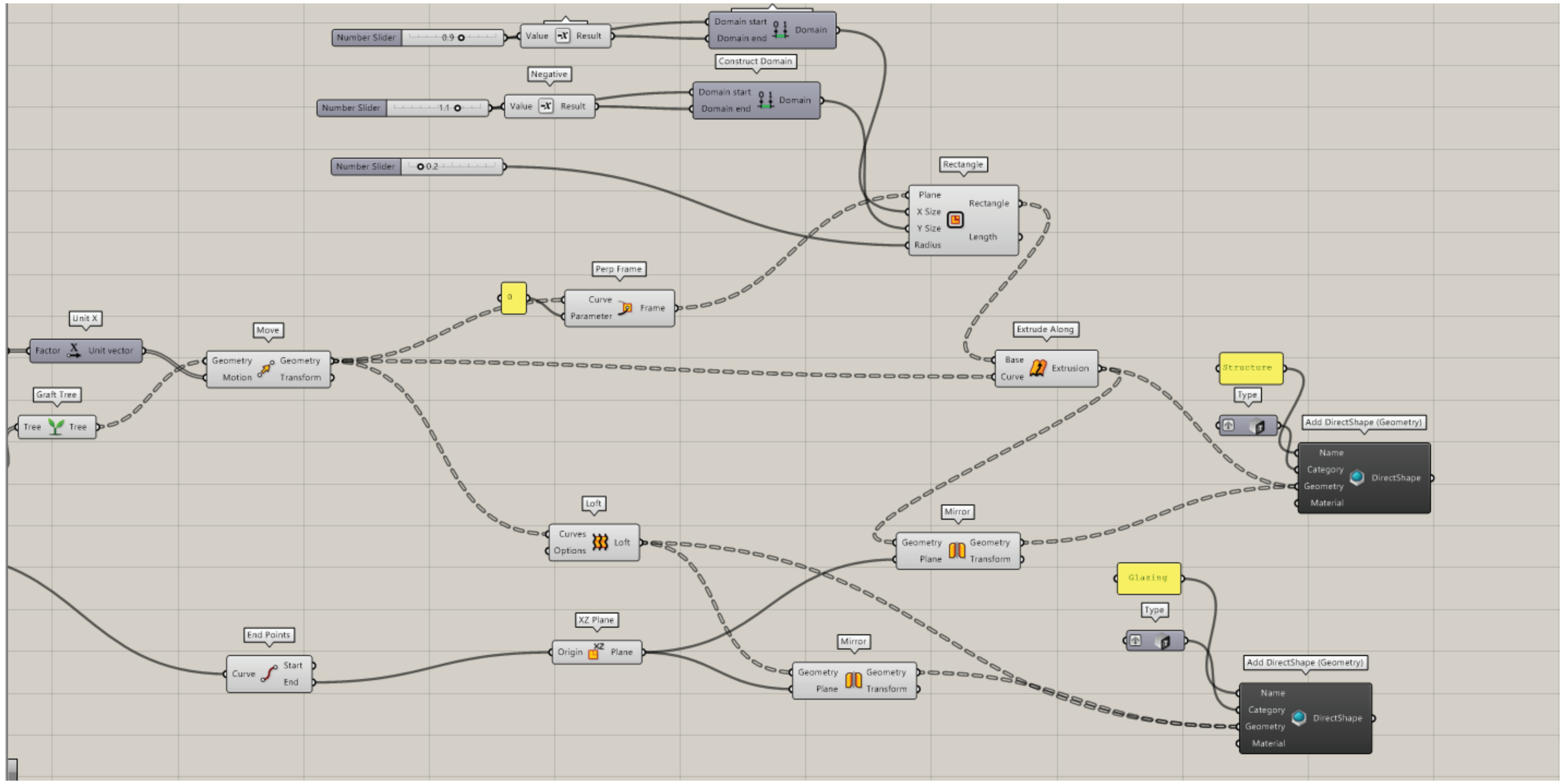
Rhino & Grasshopper

The Grasshopper script builds lines between two points, projecting the C-Plane to the top point of the first line so that the rotation may take place. This is repeated for the third line. Number sliders make the length of these segments parametric.



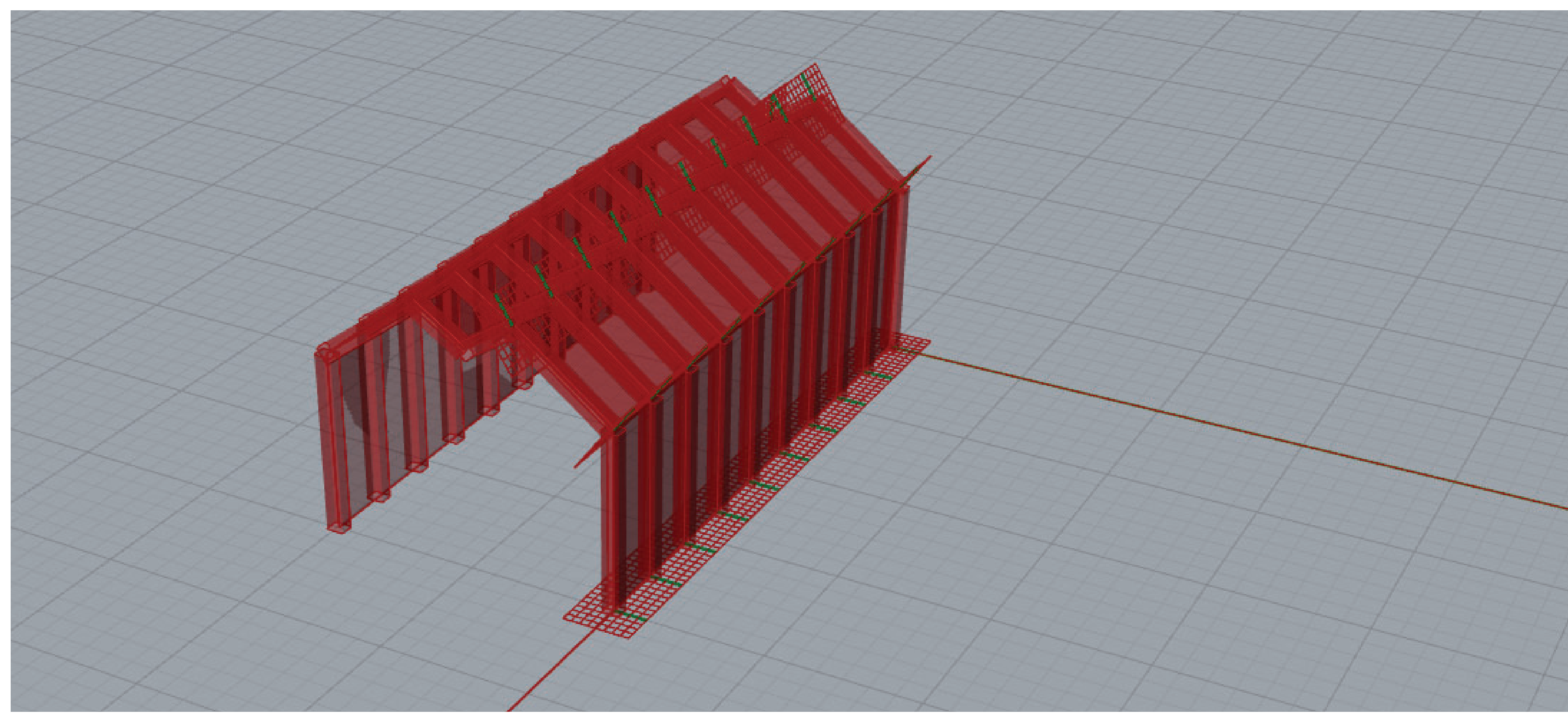
Taking the lines created a series in implemented so that multiple can be created in a single line.





Between the series if lines, a loft is implemented to create walls between each member of the series created along with adding "perp frame" that can then be imputed into a rectangle command to create the columns and beams. From there the created series is mirrored to have a complete structure that is in unison when the number sliders adjust the size or the amount the series.

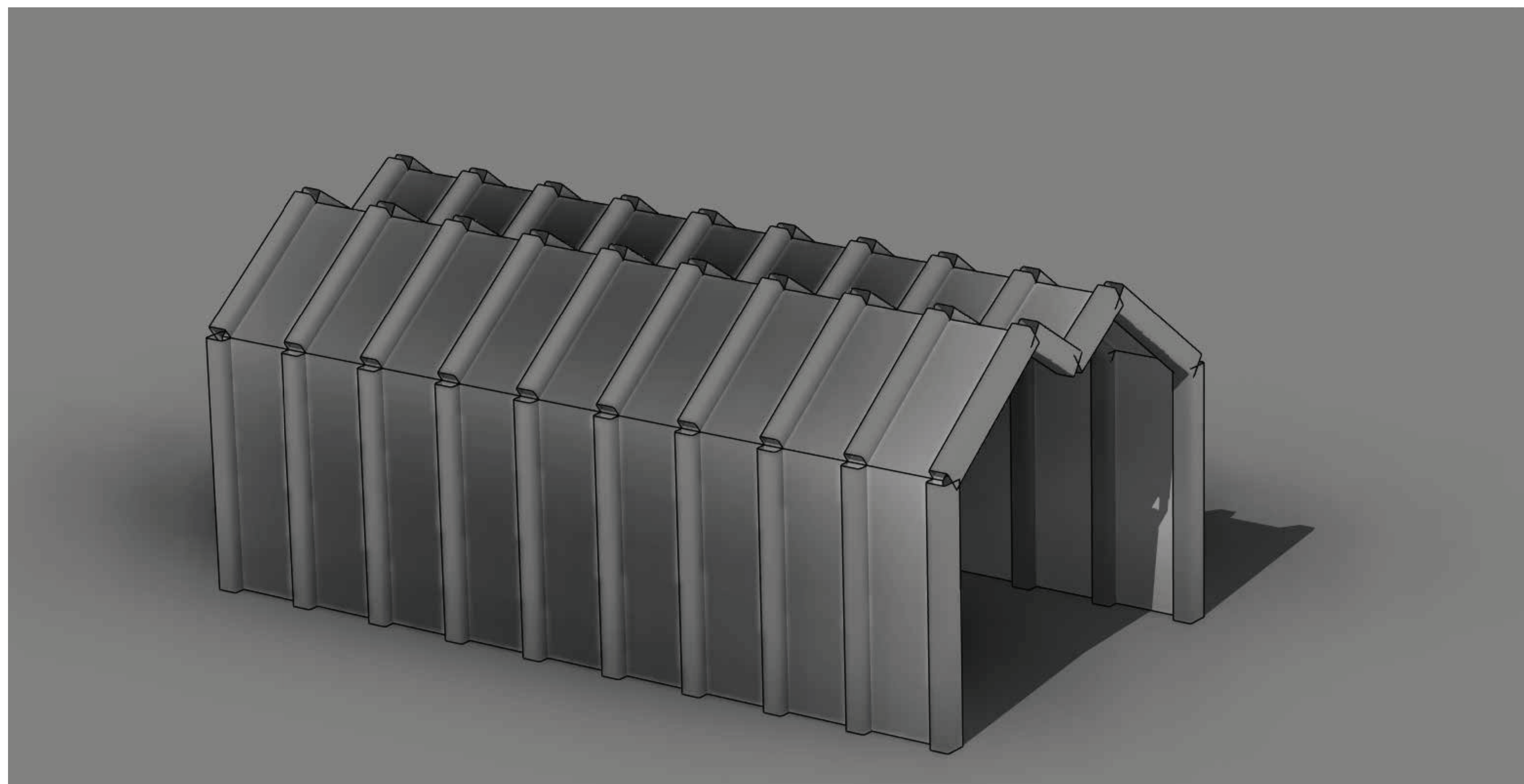
From this point it can simply be brought into Revit using direct geometry nodes to import the design with beams and walls of glazing.



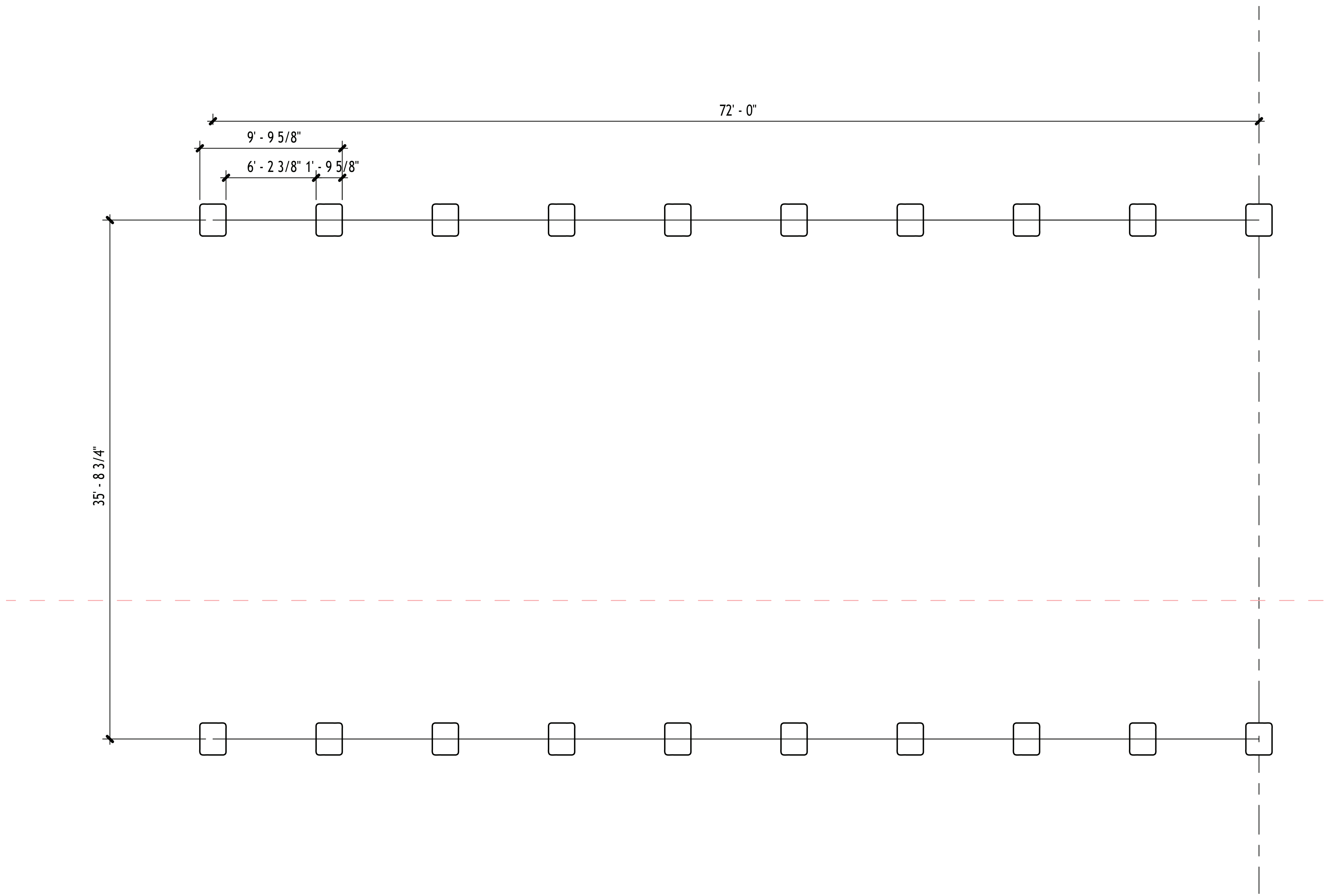
Revit

Bringing the Grasshopper script into Revit the wall were categorized as glazing and the structure was assigned wood beams.

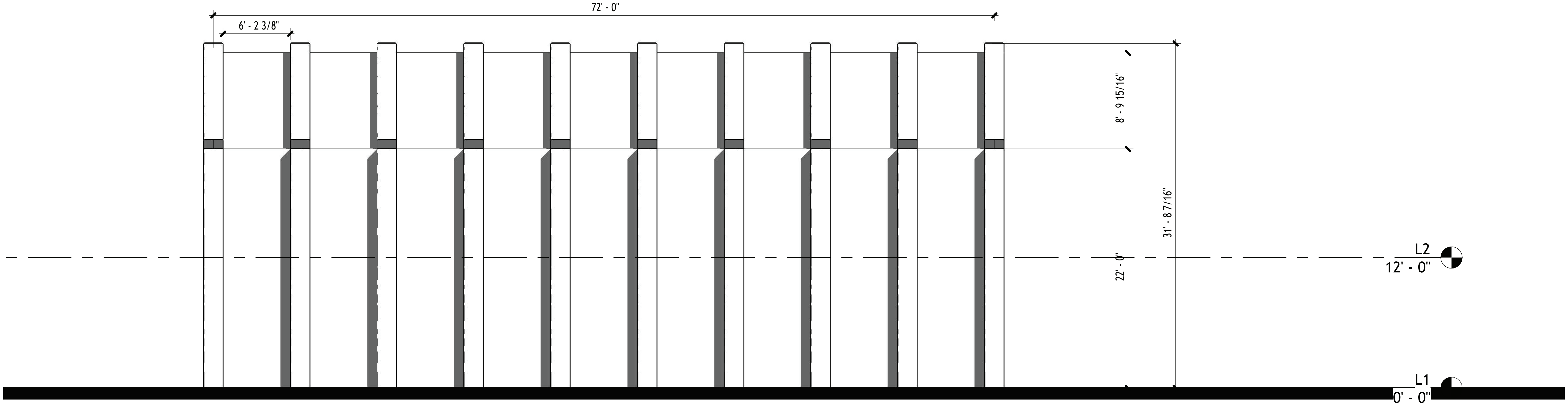
In the 3d view everything came in grey and the glazing appears to not be clear



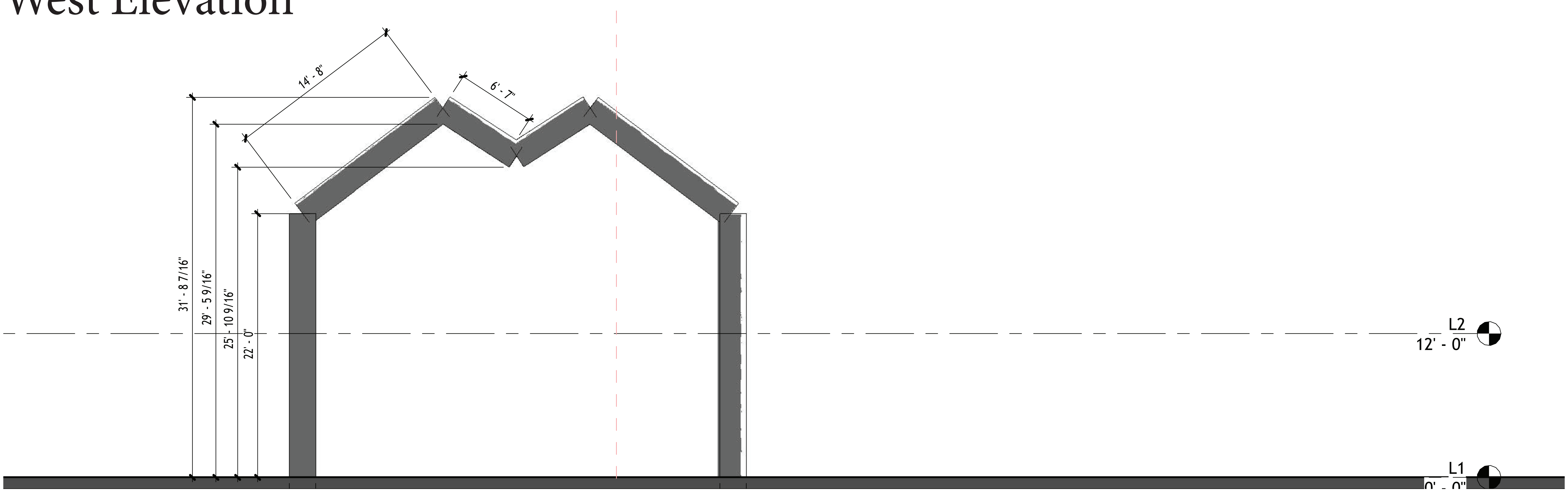
Floor Plan



South Elevation



West Elevation



Rendering

Using Twinmotion, I took my studio project into the program, rather than the parametric design made in Grasshopper, for making renders. The project is a cafe and gallery space in an undesigned site so populating the environment in a vegetative environment was best. Due to the RAM usage the exterior renders are not as populated as I would desire, so a higher end desktop would be needed for that to be achieved.